

# DIRECTORY OF OPEN-SOURCE LLMs & SLMs

Complete Landscape Guide: Frontier Open-Weights, Small Language Models, Specialized Architectures & Deployment Profiles

LLMs (7B - 671B) SLMs (0.5B - 4B) MoE & Dense Architectures Apache 2.0 / MIT / Open-Weight Licenses

**Executive Perspective:** Open-source and open-weights AI models have achieved parity with leading proprietary frontier models across natural language reasoning, code synthesis, math problem solving, and multimodal perception. Furthermore, advances in data curation (synthetic generation, curriculum filtering) have unlocked unprecedented capability in *Small Language Models (SLMs)* capable of real-time execution directly on edge hardware, smartphones, and local workstations.

## 1. Open Source Licensing & Architectural Taxonomy

When categorizing "open" models, the industry distinguishes between distinct levels of openness, architecture styles, and parameter classes:

### Open Source vs. Open Weights

- Fully Open Source (OSI-compliant):** Weights, pretraining datasets, data curation pipelines, tokenizer code, and training checkpoints are published under unrestricted licenses (e.g., Apache 2.0, MIT). *Examples: AllenAI OLMo, Hugging Face SmoLLM.*
- Open Weights / Community License:** Model weights and inference code are freely accessible for download and self-hosting, but subject to commercial user caps, safety clauses, or synthetic data restrictions. *Examples: Meta Llama 3.3, Mistral, Google Gemma 2.*

### Dense vs. Mixture-of-Experts (MoE)

- Dense Models:** Every parameter is activated for every incoming token. Simpler to quantize and serve with minimal VRAM routing overhead (e.g., Llama 3.3 70B, Qwen 2.5 72B).
- Sparse MoE:** Features a high total parameter capacity while activating only a subset per token (e.g., DeepSeek-V3 with 671B total / 37B active; Mixtral 8x22B). Delivers frontier-grade reasoning at high token generation throughput.

## 2. Frontier & Enterprise Open-Weights LLMs (High-Capacity)

These flagship open-weights models are designed for data center deployment, enterprise agent orchestration, advanced coding, and complex mathematical reasoning.

### Llama 3.1 & 3.3 Series Meta AI LLAMA 3.3 LICENSE DENSE

**Parameter Sizes:** 8B, 70B, 405B (Llama 3.1) • 70B (Llama 3.3) | **Context Window:** 128,000 tokens | **Vocab:** 128k tiktoken

The standard bearer of enterprise open-weights. Llama 3.3 70B matches the performance of the earlier 405B flagship model across major industry benchmarks (MMLU, HumanEval, Math) at a fraction of the serving cost. Pretrained on over 15 trillion multilingual tokens, natively supporting sophisticated function calling, tool use, and 128k context handling.

### DeepSeek-V3 & DeepSeek-R1 DeepSeek AI MIT LICENSE SPARSE MOE

**Parameters:** 671B Total (37B active per token) | **Context Window:** 128,000 tokens | **Architecture:** Multi-head Latent Attention (MLA)

A monumental milestone in open AI. DeepSeek-V3 provides state-of-the-art generalist capability using Multi-head Latent Attention (MLA) and DeepSeekMoE architectures. DeepSeek-R1 incorporates large-scale reinforcement learning (RL) without supervised warmup to produce superhuman chain-of-thought reasoning, rivaling OpenAI o1 on AIME, MATH-500, and Codeforces, released fully under permissive MIT.

## Qwen 2.5 Suite Alibaba Cloud APACHE 2.0 DENSE & MOE

**Parameters:** 0.5B, 1.5B, 3B, 7B, 14B, 32B, 72B • **MoE:** 57B-A14B | **Context:** Up to 128k (native), 8k output | **Languages:** 29+

Pretrained on 18 trillion multilingual tokens. Qwen 2.5 72B Instruct sets the gold standard for dense open-source models, outperforming many proprietary APIs in multilingual tasks, coding, and structural JSON extraction. Available under Apache 2.0 (up to 32B) and Qwen License for 72B. Includes specialized derivatives: Qwen2.5-Coder and QwQ reasoning model.

## Mistral Large 2 & Mixtral 8x22B Mistral AI MISTRAL RESEARCH / APACHE 2.0 MOE & DENSE

**Parameters:** Mistral Large 2 (123B Dense), Mixtral 8x22B (141B Sparse MoE / 39B active), NeMo (12B) | **Context:** 128k tokens

Pioneers of high-efficiency MoE architectures. Mistral Large 2 excels at multi-hop reasoning, complex code generation, and multi-turn enterprise workflows. Mixtral 8x22B delivers unmatched cost-per-token efficiency for high-concurrency self-hosted deployments.

## Gemma 2 Family Google DeepMind GEMMA TERMS DENSE (SLIDING WINDOW)

**Parameters:** 2B, 9B, 27B | **Context:** 8k tokens | **Technique:** Alternating sliding window attention, Logit soft-capping

Built from the same research foundations as Google's Gemini models. Gemma 2 27B punch well above its weight class, routinely outscoring 70B class models on chatbot arenas and comprehension benchmarks through advanced knowledge distillation.

## Command R+ & Aya Expanse Cohere For AI CC-BY-NC / OPEN DENSE

**Parameters:** Command R+ (104B), Command R (35B), Aya Expanse (8B, 32B) | **Context:** 128k tokens

Specially optimized for Retrieval-Augmented Generation (RAG), citation grounding, and automated enterprise workflows across 23+ operational languages. Aya Expanse expands sovereign and low-resource multilingual capabilities globally.

# 3. The Small Language Model (SLM) Landscape (< 4 Billion Parameters)

SLMs represent a massive paradigm shift: deploying highly competent intelligence locally on mobile devices, laptops (Apple Silicon / Snapdragon X Elite), edge routers, and embedded robotics with minimal power draw and memory footprint.

### Microsoft Phi-3.5 & Phi-4

**Creator:** Microsoft Research • **Size:** 3.8B (Phi-3.5 Mini), 14B (Phi-4)

**License:** MIT License

Trained extensively on "Textbooks Are All You Need" synthetic datasets. Phi-3.5 Mini achieves mathematical and reasoning scores that rival earlier 70B models while running smoothly on smartphones and single consumer GPUs.

### Meta Llama 3.2 (1B & 3B)

**Creator:** Meta AI • **Size:** 1.2B and 3.2B

**License:** Llama 3.2 Community License

Engineered explicitly for on-device edge computing, mobile text summarization, prompt rewriting, and local private tool invocation. Features 128k context and is optimized for Qualcomm NPU and Arm platforms.

### Hugging Face SmoLLM & SmoLLM2

**Creator:** Hugging Face • **Size:** 135M, 360M, 1.7B

**License:** Apache 2.0 (Fully Open)

Trained on meticulously curated synthetic textbooks and filtered web datasets (SmoLLM-Corpus). The 1.7B variant beats out many legacy 7B models on GSM8K and HumanEval, making it ideal for edge WebAssembly / browser AI.

### OpenBMB MiniCPM 3 & MiniCPM-V

**Creator:** OpenBMB / ModelBest • **Size:** 2.4B - 4B

**License:** Apache 2.0

Remarkable parameter efficiency. MiniCPM 3 4B achieves performance comparable to 8B/9B models. MiniCPM-V sets state-of-the-art multimodal vision metrics for on-device OCR and document inspection.

**Qwen 2.5 (0.5B, 1.5B, 3B)**

**Creator:** Alibaba Cloud • **Size:** 0.5B, 1.5B, 3.2B

**License:** Apache 2.0

Extreme versatility at the tiny scale. Even the sub-billion 0.5B model retains strong multilingual comprehension and structured JSON output capability, running at >100 tokens/sec on standard CPUs.

**Apple OpenELM & TinyLlama**

**Creator:** Apple Research / Open Source Community  
• **Size:** 270M - 1.1B

**License:** Sample Code License / Apache 2.0

Pioneering architectures in layer-wise parameter allocation (OpenELM) and extreme compact pretraining on 3T tokens (TinyLlama 1.1B), ideal for real-time autocomplete and IoT microcontrollers.

**4. Specialized Domain Architectures (Code, Reasoning, Vision)**

| Domain & Model         | Developer            | Sizes                    | License           | Core Capabilities & Focus                                                                                                       |
|------------------------|----------------------|--------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| DeepSeek-Coder-V2      | DeepSeek AI          | 16B, 236B MoE            | MIT               | Supports 338 programming languages; competitive with Claude 3.5 Sonnet and GPT-4o in competitive programming.                   |
| Qwen2.5-Coder          | Alibaba Cloud        | 0.5B, 1.5B, 7B, 14B, 32B | Apache 2.0        | The premier open-source code completion and editing model series; 32B matches proprietary frontier coding assistants.           |
| StarCoder 2            | BigCode / ServiceNow | 3B, 7B, 15B              | OpenRAIL-M        | Trained on The Stack v2; fully compliant software provenance, attribution, and enterprise copyright safety.                     |
| DeepSeek-R1 & QwQ-32B  | DeepSeek / Alibaba   | 32B, 671B MoE            | MIT / Apache 2.0  | Specialized test-time reasoning and dynamic chain-of-thought verification for Olympiad mathematics, logic, and proof systems.   |
| Qwen2-VL & Pixtral 12B | Alibaba / Mistral    | 2B, 7B, 72B & 12B        | Apache 2.0        | High-resolution image understanding, chart reading, video token ingestion, and graphical UI automated agent workflows.          |
| Llama 3.2 Vision       | Meta AI              | 11B, 90B                 | Llama 3.2 License | Native cross-attention image encoder integrated into Llama text backbone for document extraction and visual question answering. |
| AllenAI OLMo & OLMoE   | Allen Inst. for AI   | 1B, 7B, 1B-7B MoE        | Apache 2.0        | 100% open science: weights, code, Dolma dataset, training logs, and evaluations are fully reproducible and public.              |

**5. Master Comparison Matrix of Major Open Model Families**

The matrix below summarizes the primary open-source and open-weights model families, architectural specifications, and target hardware environments.

| Model Series | Organization | Active Sizes | Architecture      | Context | License   | Min. VRAM (Quantized) |
|--------------|--------------|--------------|-------------------|---------|-----------|-----------------------|
| Llama 3.3    | Meta AI      | 70B          | Dense Transformer | 128k    | Llama 3.3 | ~40 GB (Q4_K_M)       |
| Llama 3.2    | Meta AI      | 1B, 3B       | Dense Transformer | 128k    | Llama 3.2 | ~1.5 GB - 3 GB (Q4)   |

| Model Series        | Organization       | Active Sizes                | Architecture        | Context    | License       | Min. VRAM (Quantized)  |
|---------------------|--------------------|-----------------------------|---------------------|------------|---------------|------------------------|
| DeepSeek-V3         | DeepSeek AI        | 671B (37B active)           | Sparse MoE (MLA)    | 128k       | MIT           | ~360 GB (FP8 / NVLink) |
| DeepSeek-R1         | DeepSeek AI        | 671B (37B active)           | Reasoning MoE       | 128k       | MIT           | ~360 GB (FP8)          |
| DeepSeek-R1-Distill | DeepSeek AI        | 1.5B, 7B, 8B, 14B, 32B, 70B | Dense (Qwen/ Llama) | 128k       | MIT / Apache  | 2 GB - 42 GB (Q4)      |
| Qwen 2.5            | Alibaba Cloud      | 0.5B to 72B                 | Dense & MoE         | 128k       | Apache / Qwen | 1 GB to 42 GB (Q4)     |
| Mistral Large 2     | Mistral AI         | 123B                        | Dense Transformer   | 128k       | Mistral Comm. | ~70 GB (Q4_K_M)        |
| Mixtral 8x22B       | Mistral AI         | 141B (39B active)           | Sparse MoE          | 64k        | Apache 2.0    | ~80 GB (Q4_K_M)        |
| Gemma 2             | Google DeepMind    | 2B, 9B, 27B                 | Dense (Distilled)   | 8k         | Gemma License | 2 GB to 16 GB (Q4)     |
| Phi-3.5 / Phi-4     | Microsoft Research | 3.8B, 14B                   | Dense (Synthetic)   | 128k / 16k | MIT           | 3 GB to 9 GB (Q4)      |
| SmolLM2             | Hugging Face       | 135M, 360M, 1.7B            | Dense Edge SLM      | 8k         | Apache 2.0    | < 1.5 GB (FP16/Q4)     |
| MiniCPM 3           | OpenBMB            | 4B                          | Dense (GQA)         | 32k        | Apache 2.0    | ~3 GB (Q4)             |
| Command R+          | Cohere For AI      | 104B                        | Dense (RAG-tuned)   | 128k       | CC-BY-NC      | ~60 GB (Q4_K_M)        |
| Falcon 2 / 180B     | TII UAE            | 11B, 180B                   | Dense & VLM         | 8k         | TII Falcon    | 8 GB / 100 GB (Q4)     |
| OLMo 7B / 13B       | AllenAI            | 7B, 13B                     | Fully Open Science  | 4k         | Apache 2.0    | ~5 GB (Q4)             |
| Solar Pro / 10.7B   | Upstage AI         | 10.7B, 22B                  | Depth-Upscaled DUS  | 4k - 64k   | Apache 2.0    | ~7 GB - 14 GB (Q4)     |

## 6. Practical Hardware & Quantization Sizing Guide

Modern quantization formats (such as **GGUF**, **AWQ**, **EXL2**, and native **FP8**) permit running high-parameter models on modest hardware with negligible degradation in reasoning accuracy.

### VRAM Footprint Rule of Thumb

- **16-bit (FP16/BF16):** ~2.0 GB VRAM per 1 Billion parameters + KV cache.
- **8-bit (INT8 / FP8):** ~1.1 GB VRAM per 1 Billion parameters + KV cache.
- **4-bit (INT4 / Q4\_K\_M):** ~0.65 GB to 0.75 GB VRAM per 1 Billion parameters.
- *Example:* A 70B parameter model in 4-bit requires ~45 GB VRAM (fits on 2× RTX 3090/4090 or single 64GB+ Apple Silicon Mac).

### Recommended Serving Frameworks

- **vLLM & SGLang:** High-throughput production enterprise serving with PagedAttention, continuous batching, and tensor parallelism.
- **Ollama:** Zero-configuration local runner for macOS, Windows, and Linux powered by llama.cpp backends.
- **llama.cpp:** Bare-metal C/C++ cross-platform inference for CPU, Apple Metal, ROCm, and CUDA.
- **TensorRT-LLM:** Ultra-low latency NVIDIA-optimized enterprise deployment.

**Summary & Selection Recommendation:** For on-device edge deployments (phones, tablets, IoT), select **Phi-3.5 Mini (3.8B)**, **Llama 3.2 (1B/3B)**, or **SmolLM2 (1.7B)**. For cost-effective enterprise production on single GPUs (16-24GB VRAM), choose **Qwen 2.5 14B/32B** or **DeepSeek-R1-Distill-Qwen-14B/32B**. For uncompromised frontier reasoning and complex code orchestration, self-host **Llama 3.3 70B**, **DeepSeek-V3**, or **DeepSeek-R1**.